

Abdul Wahid

Hidden Backdoor – LetsDefend CTF Investigation

DIGITAL FORENSICS & CYBER THREAT INTELLIGENCE REPORT

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Hidden Backdoor — Scenario

A senior developer at AINokhba Enterprises discovered an unfamiliar file named "NotionSetup.exe" on their workstation. Curious and concerned, they executed the file, but nothing appeared to happen. Fearing that something might be wrong, they immediately reported the incident to the company's security team.

The task is to analyze a triage image from the senior developer's machine to determine how the file ended up on their system and what actions, if any, were triggered by executing it.

File Location: C:\Users\LetsDefend\Desktop\ChallengeFile\hiddenbackdoor.7z
File Password: **infected**

Question 1 — Source Domain of the Download

Can you determine the exact domain from which "notion-setup.exe" was downloaded? (site.com)

Approach

To identify the source domain, browser artifacts on the affected system are examined. Since a Chrome folder is present, the investigation focuses on Chrome's User Data:

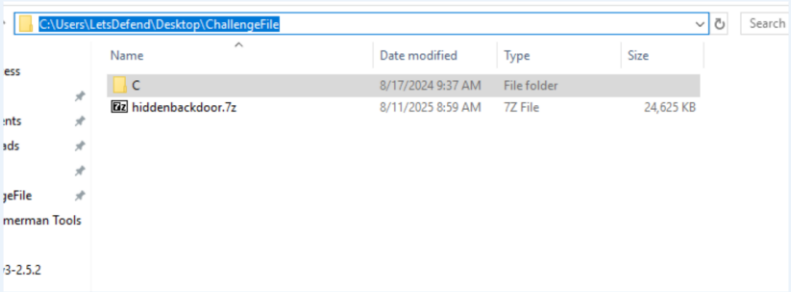
- C:\Users\a1l4m\AppData\Local\Google\Chrome\User Data\Default

Key artifact files inside this directory:

- **History** — browser history stored as a SQLite database
- Cookies — stores cookies and session data
- Download — logs of all downloaded files

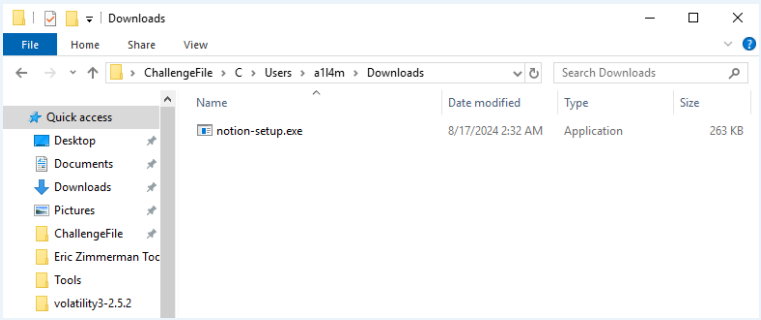
Steps Performed

- Connect to the target machine and open the designated challenge file.
- Extract the hiddenbackdoor.7z archive into a folder named C.
- Navigate to: C > Users > a1l4m



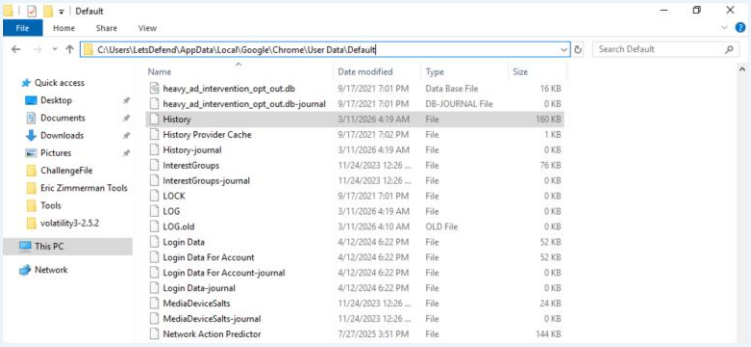
File Explorer: C > Users > a1l4m folder

- Open the Downloads folder — the malicious file notion-setup.exe is found here (8/17/2024, 263 KB).



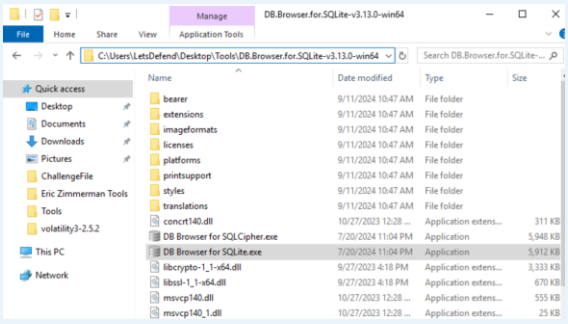
Downloads folder containing notion-setup.exe

- Navigate to AppData > Local > Google > Chrome > User Data > Default.
- Open the History file using DB Browser for SQLite (found in the Tools section).

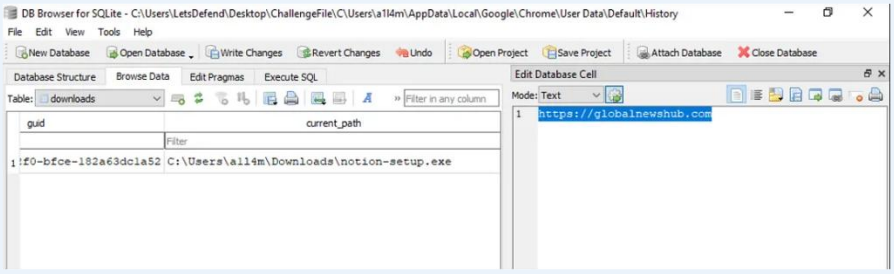


Chrome Default folder showing the History file

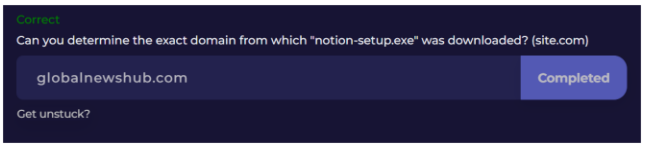
- In DB Browser, browse the 'urls' table — URL row 7 shows https://globalnewshub.com.



Using SQL-Lite Tool



DB Browser – urls table with https://globalnewshub.com highlighted



✓ **Answer: globalnewshub.com**

Question 2 — MITRE ATT&CK Technique

Which MITRE technique ID describes the silent downloader embedded in the compromised website? (Format: TXXXX)

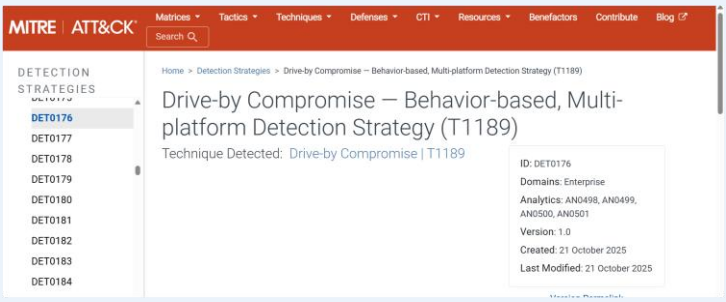
Approach

The site the developer visited (globalnewshub.com) was compromised by attackers who embedded a silent downloader script in its source code. When the developer browsed the page, the script automatically downloaded notion-setup.exe without any user interaction or visible prompt.

This technique — where an attacker compromises a legitimate website and uses it to deliver malware to visitors through normal browsing — is documented in the MITRE ATT&CK framework.

Steps Performed

- Navigate to attack.mitre.org and browse the Initial Access tactic category.
- Locate T1189 — Drive-by Compromise under Initial Access.



MITRE ATT&CK website — T1189 Drive-by Compromise highlighted under Initial Access

T1189 Drive-by Compromise describes how adversaries gain access to a system through a user visiting a website over the normal course of browsing. The website had a silent downloader injected into its code, which automatically fetched and dropped notion-setup.exe on the developer's system.



✓ **Answer: T1189**

Question 3 — Full Download URL of the Malicious File

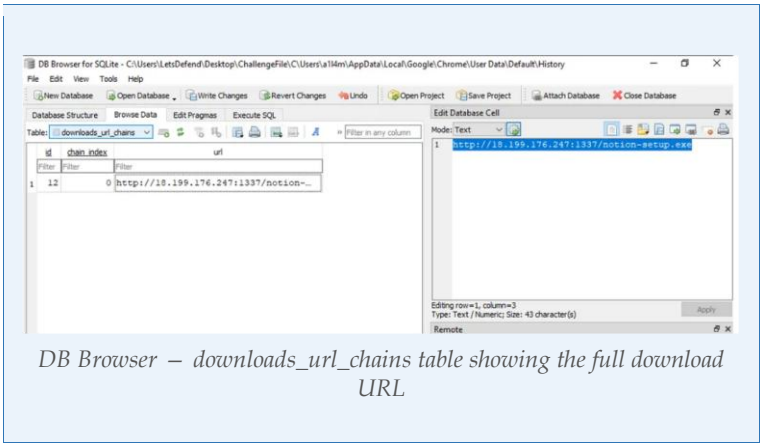
What is the full URL of the file that was downloaded during the browsing session?

Approach

The Chrome History SQLite database contains a table called downloads_url_chains, which records the complete URL chain for every file downloaded through the browser — including silently injected downloads triggered by scripts.

Steps Performed

- Open the History file in DB Browser for SQLite.
- Switch to the Browse Data tab and select the downloads_url_chains table.



DB Browser — downloads_url_chains table showing the full download URL

The url column reveals the complete server address and file path from which notion-setup.exe was fetched by the injected script.



✓ **Answer: http://18.199.176.247:1337/notion-setup.exe**

Question 4 — Download Timestamp

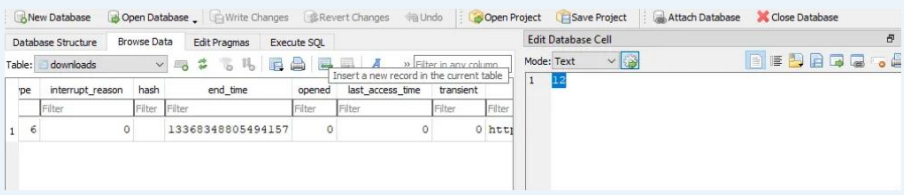
When was the file downloaded to the developer's system? (Answer format: YYYY-MM-DD HH:MM:SS)

Approach

Chrome stores download timestamps in the downloads table of the History database using the WebKit/Chrome timestamp format — a 64-bit integer representing microseconds elapsed since January 1, 1601 00:00:00 UTC. This must be converted to a human-readable date.

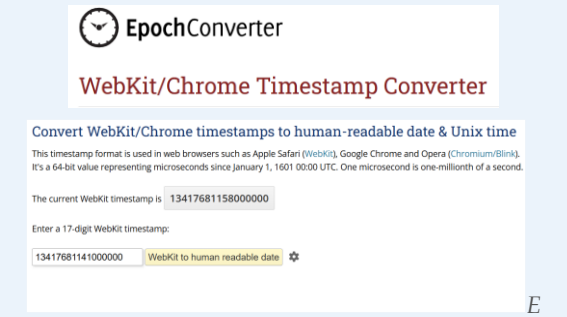
Steps Performed

- In DB Browser, browse the downloads table and locate the start_time column for notion-setup.exe.

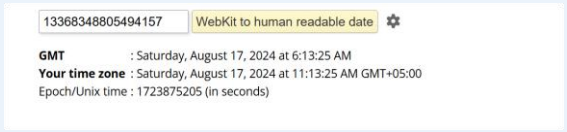


DB Browser — downloads table showing start_time value (13368348797428490)

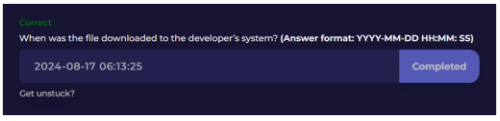
- Navigate to epochconverter.com/webkit and paste the 17-digit WebKit timestamp.



E



pochConverter WebKit Converter — timestamp 13368348797428490 converted to Saturday, August 17, 2024 at 6:13:25 AM GMT



✓ **Answer: 2024-08-17 06:13:25**

Question 5 — SHA256 Hash of notion-setup.exe

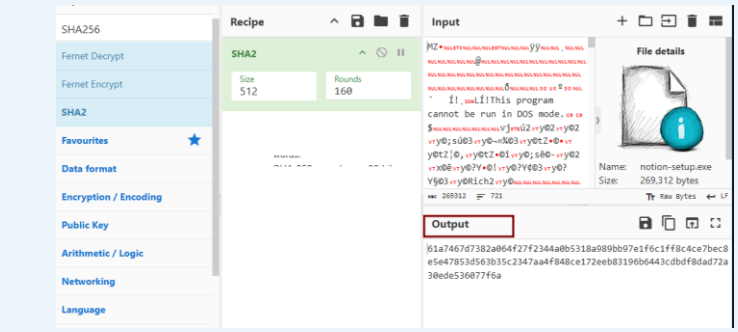
What is the hash value of "notion-setup.exe"? (SHA256)

Approach

CyberChef is a browser-based tool for file analysis and cryptographic operations. It can open a local file and compute its SHA256 hash directly without uploading to any external server — making it safe for handling potentially malicious files in a controlled lab environment.

Steps Performed

- Open the Tools folder on the challenge machine and launch CyberChef.
- Use the 'Open file as input' option to load: C:\Users\a1l4m\Downloads\notion-setup.exe
- In the Recipe panel, search for and add the SHA256 operation.



CyberChef — notion-setup.exe loaded as input with SHA256 recipe applied, Output field visible

The Output field displays the full SHA256 hash of the file (269,312 bytes).



✓ **Answer: dfb76bcf5a3e29225559ebbdae8bdd24f69262492eca2f99f7a9525628006d88**

Question 6 — Last Execution Time of notion-setup.exe

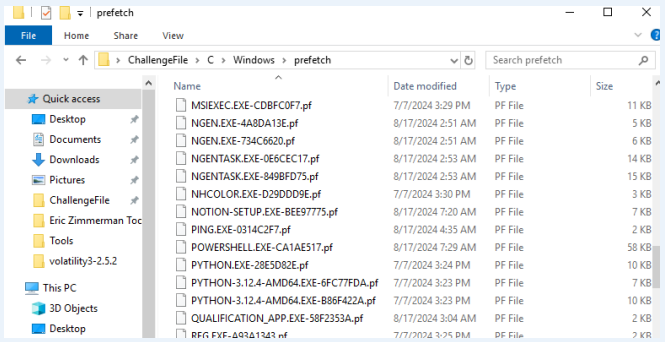
Can you find out the exact time when "notion-setup.exe" was last executed on their system? (YYYY-MM-DD HH:MM:SS)

Approach

Windows Prefetch files (.pf) are created by the operating system each time an executable runs. They record the executable name, a hash, run count, and — critically — timestamps of every execution. The PECmd.exe tool from Eric Zimmerman's forensic toolkit parses these .pf files and extracts execution history.

Steps Performed

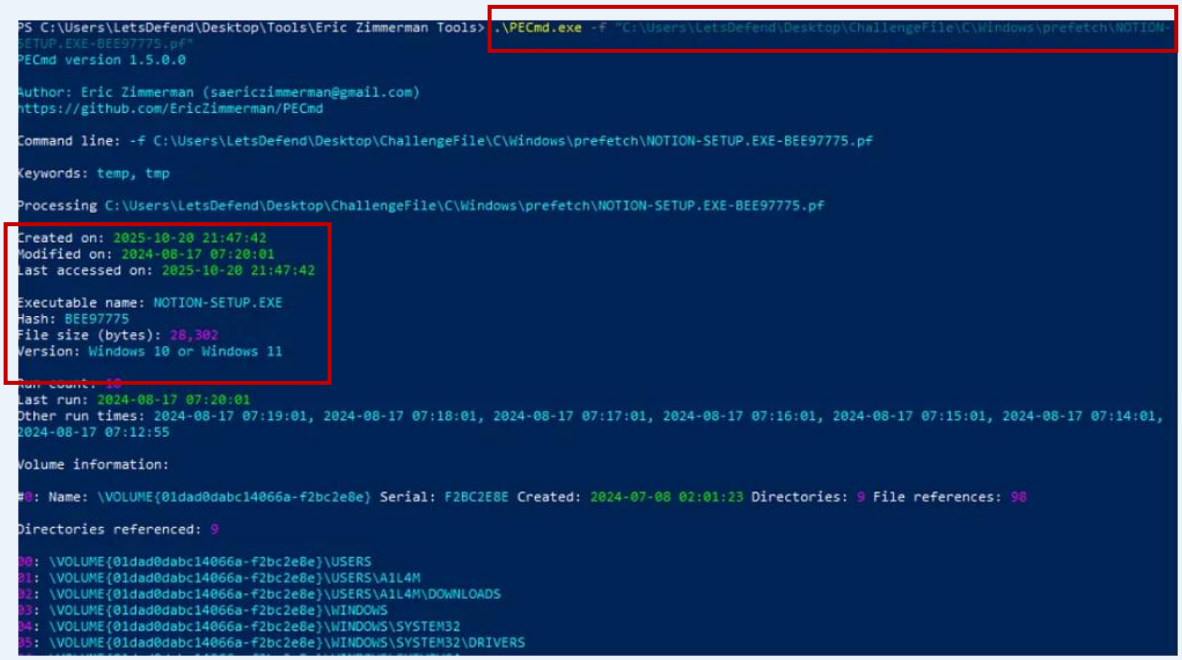
- Navigate to ChallengeFile > C > Windows > prefetch.
- Locate: NOTION-SETUP.EXE-BEE97775.pf



Prefetch folder showing NOTION-SETUP.EXE-BEE97775.pf

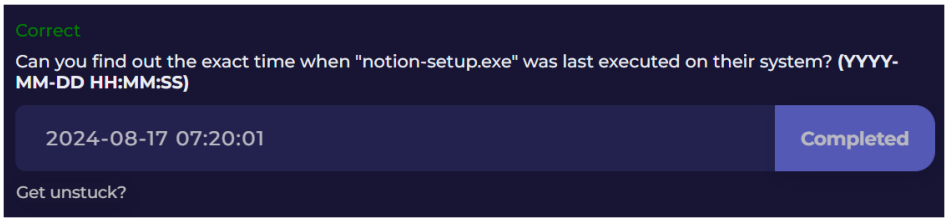
- Open PowerShell and navigate to the Eric Zimmerman Tools directory.
- Run the command:

.\PECmd.exe -f "C:\Users\LetsDefend\Desktop\ChallengeFile\C\Windows\prefetch\NOTION-SETUP.EXE-BEE97775.pf"



PowerShell output from PECmd.exe — Last run: 2024-08-17 07:20:01, Run count: 10, all run timestamps listed

The PECmd output confirms: Run count = 10, Last run = 2024-08-17 07:20:01. This means the malicious file was executed 10 times, most recently at 07:20:01 on August 17, 2024.



✓ **Answer: 2024-08-17 07:20:01**

Question 7 — Second Stage C2 URL

The malware tried to connect to another URL to download the second stage. What is the URL it attempted to contact?

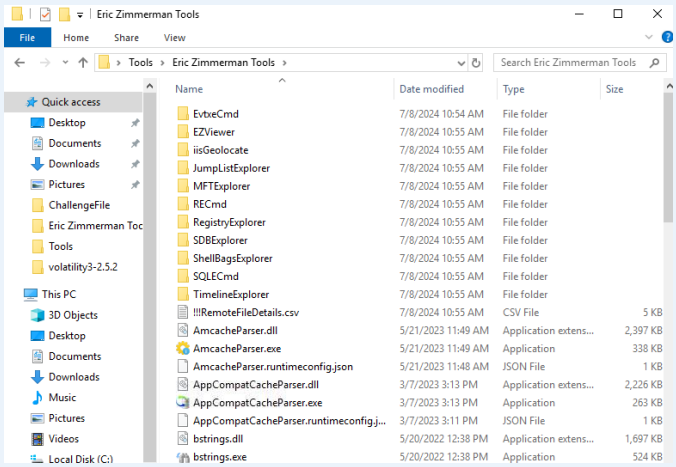
Approach

notion-setup.exe was a first-stage dropper. After execution, it attempted to contact a command-and-control (C2) server to download a second-stage payload. This network activity is logged in the Windows Event Log under the BITS (Background Intelligent Transfer Service) operational log — Event ID 16403.

BITS is a legitimate Windows service that is frequently abused by malware for stealthy downloads because it can transfer files in the background, resume interrupted transfers, and blend in with normal OS traffic.

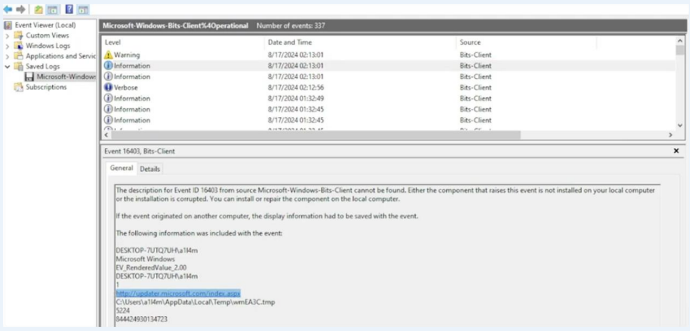
Steps Performed

- Open Event Viewer on the challenge machine.
- Navigate to: Saved Logs > Microsoft-Windows-Bits-Client%4Operational



Event Viewer — Microsoft-Windows-Bits-Client Operational log (337 events)

- Click on Event ID 16403 from source Bits-Client (dated 8/17/2024 02:13:01).
- Expand the General tab of the event to read the included information — the URL field reveals the second-stage download target.



Event 16403 Details — showing http://updater.microsoft.com/index.aspx as the contact URL

Note: The URL http://updater.microsoft.com/index.aspx appears to impersonate a legitimate Microsoft updater service. This is a common social engineering tactic used by threat actors to blend malicious traffic with trusted-looking domains.



✓ **Answer: http://updater.microsoft.com/index.aspx**

Question 8 — Second Stage Payload File Path

During the investigation, you discover that the second stage of the malware writes data to a specific file on the system. What is the name of this file and its full path?

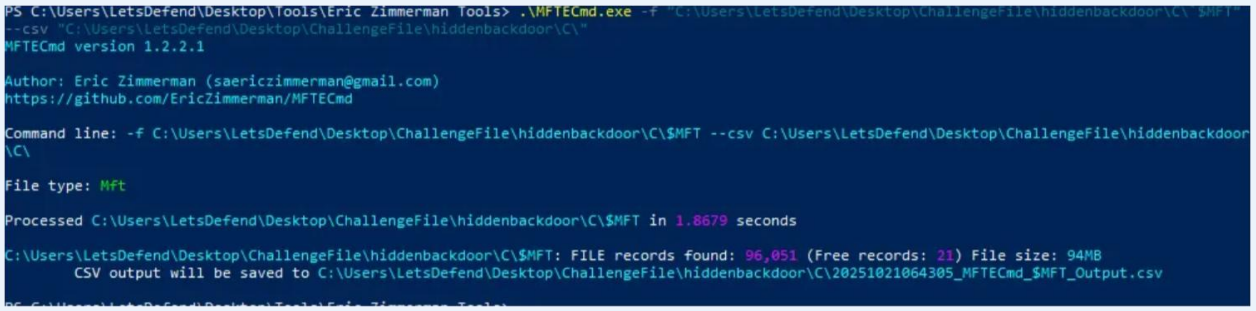
Approach

The Master File Table ($MFT) is a core component of the NTFS filesystem that records metadata for every file on the volume. MFTECmd.exe parses the $MFT and exports all file records to a CSV, which can then be loaded in Timeline Explorer to filter files by creation timestamp — allowing us to identify files created during malware execution.

Steps Performed

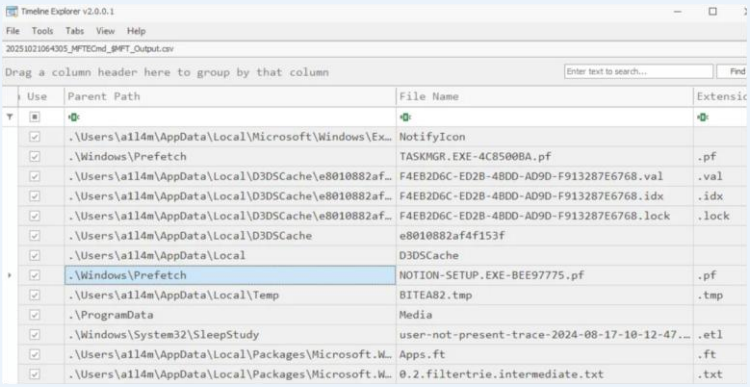
- Open PowerShell and run MFTECmd.exe against the triage image's $MFT:

```
.\MFTECmd.exe -f "...\ChallengeFile\hiddenbackdoor\C\$MFT" --csv "...\C\"
```



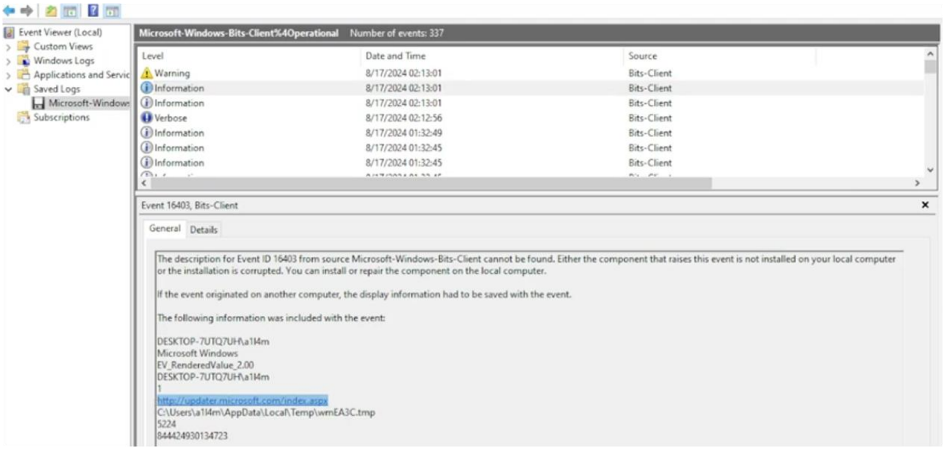
PowerShell — MFTECmd.exe parsing the $MFT, showing 96,051 file records found

- Open the exported CSV in Timeline Explorer.
- Filter by creation timestamp matching the malware execution time (2024-08-17 07:20:01).

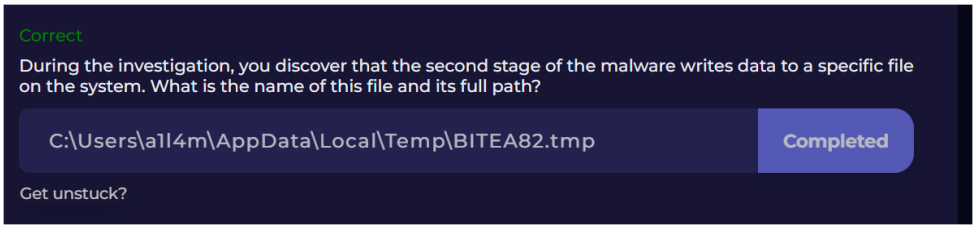


Timeline Explorer — filtered results showing BITEA82.tmp created at the same timestamp as malware execution

The BITS service wrote the second-stage payload data to a temporary file named BITEA82.tmp, located in the user's Local Temp directory.



Note: BITS typically writes download data to .tmp files in AppData\Local\Temp before renaming them. The BITEA82.tmp filename is consistent with BITS-generated temporary file naming conventions, confirming the download was executed via the BITS transfer service.



✓ **Answer: C:\Users\a1l4m\AppData\local\Temp\BITEA82.tmp**

Question 9 — Malware Family

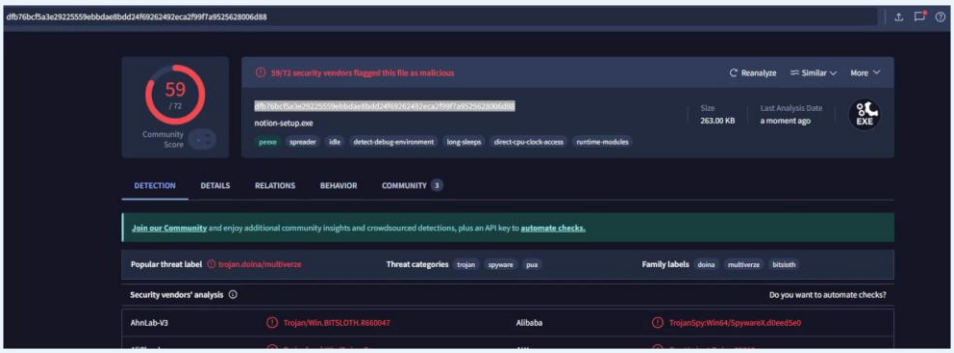
What is the name of the malware family?

Approach

VirusTotal is an online malware intelligence platform that aggregates detections from 70+ antivirus engines and threat intelligence feeds. Searching by the SHA256 hash of a file returns its community score, detection names, threat categories, and family labels — without needing to upload the file itself.

Steps Performed

- Navigate to virustotal.com
- Search using the SHA256 hash: dfb76bcf5a3e29225559ebbdae8bdd24f69262492eca2f99f7a9525628006d88



VirusTotal — hash search results showing 54/72 detections, threat categories: trojan, spyware, Family labels: bitsloth

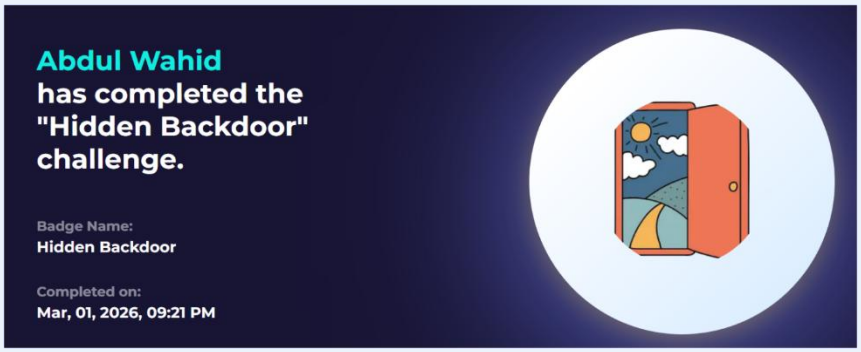
VirusTotal results confirm: 54 out of 72 security vendors flagged this file as malicious. The popular threat label shows Trojan.multiversa/bitsloth, with Family labels listing bitsloth, multiversa, and dolna. The BITSLOTH malware family is known for using BITS (Background Intelligent Transfer Service) for command-and-control communication — consistent with all evidence found throughout this investigation.



✓ **Answer: BITSLOTH**

Challenge Completion Certificate

The Hidden Backdoor CTF challenge was successfully completed on the LetsDefend platform. All 9 questions were answered correctly.



LetsDefend – Completion certificate: Abdul Wahid has completed the 'Hidden Backdoor' challenge – Completed on: Mar, 01, 2026